Precalculus II

An Original Ten-Week Course Reader

Dr. Lee MATH& 142

Contents

1 Angles, Rotation, and the Unit Circle **4**
1.1 Angles measure turning . . . 4
1.2 Why radians are natural . . . 4
1.3 Arc length and angular speed . . . 5
1.4 Rotation becomes coordinates . . . 6
1.5 Reference angles and periodicity . . . 7
1.6 Where exact coordinates come from . . . 7
1.7 From the circle to a wave . . . 7
1.8 A complete mathematical model . . . 8
1.9 Exercises . . . 8

2 The Six Trigonometric Functions **10**
2.1 From a point to six functions . . . 10
2.2 Why tangent is a slope . . . 10
2.3 Signs and the quadrants . . . 11
2.4 Where are the functions undefined? . . . 12
2.5 Zeros, poles, and disconnected graphs . . . 12
2.6 Right triangles are a special case . . . 12
2.7 Special angles without memorizing a disconnected table . . . 13
2.8 Reading the traditional trigonometric graphs . . . 13
2.9 Applications: orientation and measurement . . . 14
2.10 Measuring height and choosing the right ratio . . . 14
2.11 Exercises . . . 14

3 Periodic Functions and Their Graphs **16**
3.1 The circle makes a wave . . . 16
3.2 Amplitude and vertical translation . . . 16
3.3 Period and horizontal scaling . . . 16
3.4 Phase shift . . . 16
3.5 Five landmark points . . . 17
3.6 Models of repeating motion . . . 17
3.7 Tangent and reciprocal graphs . . . 17
3.8 Exercises . . . 17

4 Inverse Trigonometry and Identities **18**
4.1 An inverse is not a reciprocal . . . 18
4.2 Compositions and triangle geometry . . . 18
4.3 An identity is true throughout its domain . . . 18
4.4 Angle addition and double angles . . . 19
4.5 Applications: measured angles . . . 19
4.6 Exercises . . . 19

5 Trigonometric Equations and Applications **20**
5.1 A first equation on the unit circle . . . 20
5.2 Equations with transformations . . . 20
5.3 Factoring and checking domains . . . 20
5.4 Trigonometric identities in equations . . . 20
5.5 Solving triangles: notation . . . 21
5.6 The ambiguous SSA case . . . 21
5.7 Applied modeling and interpretation . . . 21
5.8 Exercises . . . 21

6 Vectors and Geometric Applications **23**
6.1 From coordinates to directed displacements . . . 23
6.2 Vector arithmetic . . . 23
6.3 Direction and trigonometry . . . 23
6.4 The dot product measures alignment . . . 24
6.5 Projection and useful components . . . 24
6.6 A geometric connection to the Law of Cosines . . . 24
6.7 Exercises . . . 24

7 Polar Coordinates and Complex Numbers **26**
7.1 Two descriptions of the same point . . . 26
7.2 Polar coordinates are not unique . . . 26
7.3 Complex numbers as plane vectors . . . 26
7.4 Addition and multiplication mean different geometries . . . 27
7.5 De Moivre's theorem and roots . . . 27
7.6 Polar graphs: a different way to draw a curve . . . 27
7.7 Exercises . . . 27

8 Parametric Equations and Conic Sections **29**
8.1 A moving point traces a curve . . . 29
8.2 Eliminating the parameter . . . 29
8.3 Projectile motion as a parametric curve . . . 29
8.4 Conics: circles and ellipses . . . 30
8.5 Parabolas and their focus . . . 30
8.6 Hyperbolas and differences of distances . . . 30
8.7 Changing descriptions . . . 30
8.8 Exercises . . . 30

9 Sequences and Series **32**
9.1 A sequence is an ordered list . . . 32
9.2 Arithmetic sequences: repeated addition . . . 32
9.3 Geometric sequences: repeated multiplication . . . 32
9.4 Sigma notation . . . 33
9.5 Infinite geometric series . . . 33
9.6 Recursive rules beyond arithmetic and geometric patterns . . . 33
9.7 Exercises . . . 33

10 Synthesis: Motion and Patterns **34**
10.1 One motion, several descriptions . . . 34
10.2 Velocity, vectors, and periodicity . . . 34
10.3 Sampling creates a sequence . . . 34
10.4 Scaling and rotation together . . . 35
10.5 Sums represent accumulated effects . . . 35
10.6 Different curves arise from different constraints . . . 35
10.7 A complete example . . . 35
10.8 Exercises . . . 35

1 Angles, Rotation, and the Unit Circle

This week's central question: How can we turn rotation into coordinates? The answer begins with the natural geometry of a circle.

The first half of precalculus introduced functions as ways of turning inputs into outputs. In this course, an angle is the input, and the outputs will be coordinates, distances, and measurements of motion. The geometry is worth understanding before trying to memorize a table of values. A circle gives us a concrete place to see what every formula is describing.

Learning goals. By the end of the chapter, you should be able to describe signed rotation, convert between degrees and radians, calculate arc length and angular speed, locate unit-circle points exactly, and explain how a turning point produces two repeating coordinate functions.

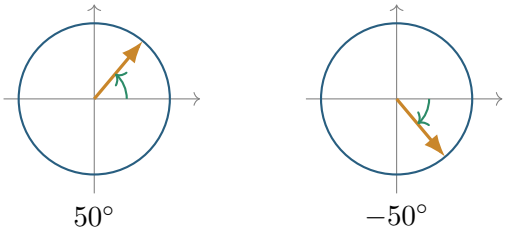
1.1 Angles measure turning

An angle records a rotation from one ray to another. Counterclockwise rotations are positive, and clockwise rotations are negative. A rotation can exceed one full revolution. For example, 450° and 90° have the same terminal ray.

Definition 1.1. Two angles are *coterminal* if they have the same initial and terminal rays. They differ by an integer multiple of 360°, or 2π radians.

Example 1.1. A positive coterminal angle for $-50°$ is $-50° + 360° = 310°$.

To make the meaning of a signed angle visible, imagine an arrow anchored at the origin. Its initial position points right. A positive angle turns the arrow counterclockwise; a negative angle turns it clockwise. The angle records the *turning*, not simply the final direction. Thus 30°, 390°, and −330° point along the same terminal ray, although their rotations are different.



Worked example: multiple turns. Find an angle between 0° and 360° coterminal with −770°. Adding 360° twice gives −50°; adding it once more gives 310°. This is the terminal direction even though the original rotation made more than two clockwise turns.

Try it. Sketch −120°, 240°, and 600°. Which terminal rays coincide? How can you tell without drawing all of the turns?

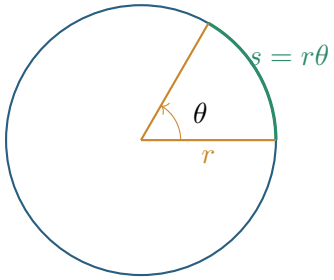
1.2 Why radians are natural

For a circle of radius r with intercepted arc length s, define the radian measure $\theta = s/r$. This is a ratio of lengths. A full circumference is $2\pi r$, so one revolution measures 2π radians:

$$\boxed{\theta = \frac{s}{r}}, \qquad 180° = \pi \text{ radians}.$$

Example 1.2. Convert 150° to radians: $150(\pi/180) = 5\pi/6$. Convert $7\pi/4$ radians to degrees: $(7\pi/4)(180/\pi) = 315^\circ$.

Try it. Convert 225° to radians and $-\pi/3$ radians to degrees, and sketch their terminal rays.



Angle measured by the arc it intercepts

The radian definition has an important consequence: the radius cancels when we compare the corresponding arc length with the radius. A 60° turn subtends a longer arc on a large circle than on a small one, but the ratio s/r is the same. That is why the angle itself is independent of the circle's size.

Degrees	Radians	Fraction of one turn
30°	$\pi/6$	$1/12$
45°	$\pi/4$	$1/8$
60°	$\pi/3$	$1/6$
90°	$\pi/2$	$1/4$
180°	π	$1/2$
270°	$3\pi/2$	$3/4$
360°	2π	1

The table is not a second set of unrelated facts to memorize. Every entry comes from the single equality $180^\circ = \pi$ radians. For instance, 60° is one third of 180°, so it is $\pi/3$ radians.

Example 1.3. Convert a negative angle: $-210^\circ \cdot \pi/180^\circ = -7\pi/6$. Its terminal ray coincides with $5\pi/6$, because adding 2π to $-7\pi/6$ gives $5\pi/6$.

1.3 Arc length and angular speed

Rearrange the definition of radian measure to obtain $s = r\theta$. If rotation through θ takes time t, angular speed is $\omega = \theta/t$ and tangential speed is $v = r\omega$.

Example 1.4. A wheel of radius 0.4 m turns through $3\pi/2$ radians. A point on its rim travels $s = 0.4(3\pi/2) = 0.6\pi$ m.

Try it. A Ferris wheel of radius 12 m makes one revolution every 40 s. Find its angular speed and a passenger's tangential speed.

Worked example: a carousel. A carousel makes one revolution every 24 seconds. A child sits 3 meters from the axis, while another sits 5 meters from the axis. Both complete the same angular displacement in the same amount of time, so their angular speed is

$$\omega = \frac{2\pi}{24} = \frac{\pi}{12} \text{ rad/s}.$$

But their speeds along their respective circles differ:

$$v_1 = 3\frac{\pi}{12} = \frac{\pi}{4} \text{ m/s}, \qquad v_2 = 5\frac{\pi}{12} = \frac{5\pi}{12} \text{ m/s}.$$

The outer rider travels farther in the same time. This is the distinction between *angular speed* and *tangential speed*.

A caution about units. The formula $s = r\theta$ requires θ in radians. Substituting 90 for a 90° angle would calculate ninety radii of arc instead of a quarter-circumference. First convert 90° to $\pi/2$.

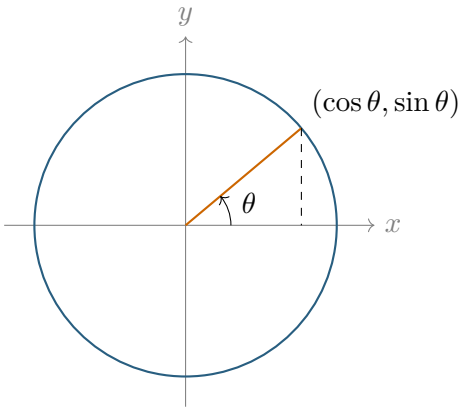
Try it. A wheel of radius 0.6 m turns through 135° in 3 seconds. Find the arc length, average angular speed, and average tangential speed. Explain the units of all three answers.

1.4 Rotation becomes coordinates

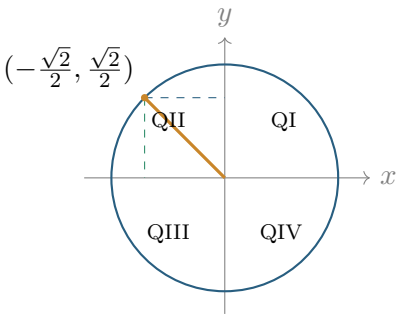
Take a circle of radius 1 centered at the origin. The point at angle θ, measured from the positive x-axis, has coordinates

$$\boxed{(x, y) = (\cos\theta, \sin\theta)}.$$

Cosine is the horizontal coordinate and sine is the vertical coordinate. Because the point lies on $x^2 + y^2 = 1$, we get $\cos^2\theta + \sin^2\theta = 1$.



We can now make the coordinate interpretation precise. Begin with the point $(1, 0)$ and rotate it through θ. The new point is $(\cos\theta, \sin\theta)$. The horizontal coordinate can be negative, and so can the vertical coordinate: the point is not confined to a triangle in the first quadrant.



Worked example: a point in Quadrant III. At $225° = 5\pi/4$, the reference angle is $\pi/4$. The two coordinates have equal absolute value, but both are negative:

$$(\cos(5\pi/4), \sin(5\pi/4)) = \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right).$$

This example illustrates why taking the positive side lengths of a reference triangle is only the first step. The signs come from the location in the plane.

Try it. Without a calculator, describe the point on the unit circle at 150° and at 330°. Which coordinates have the same absolute value? Which signs differ?

1.5 Reference angles and periodicity

Special angles arise from right triangles:

θ	$\cos\theta$	$\sin\theta$
0	1	0
$\pi/6$	$\sqrt{3}/2$	$1/2$
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$
$\pi/3$	$1/2$	$\sqrt{3}/2$
$\pi/2$	0	1

The reference angle supplies the absolute values in every quadrant; the quadrant gives the signs.

Example 1.5. At $5\pi/6$, the reference angle is $\pi/6$ and the point is in quadrant II, so $\cos(5\pi/6) = -\sqrt{3}/2$ and $\sin(5\pi/6) = 1/2$.

After one full revolution the point returns to its previous position:

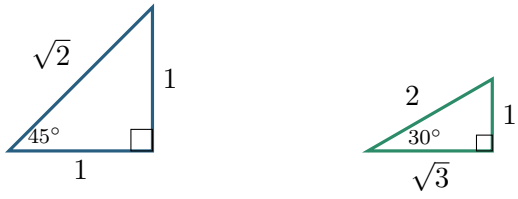
$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

This geometric periodicity will become the foundation of sinusoidal graphs.

1.6 Where exact coordinates come from

Instead of treating the exact-value table as a memorization task, we can derive its entries from two familiar triangles.

An isosceles right triangle with legs 1 has hypotenuse $\sqrt{2}$. Dividing both legs by $\sqrt{2}$ yields the point $(\sqrt{2}/2, \sqrt{2}/2)$ at 45°. Likewise, a 30°–60°–90° triangle has side lengths in the ratio $1 : \sqrt{3} : 2$. Dividing by its hypotenuse produces the coordinates $(\sqrt{3}/2, 1/2)$ for 30° and $(1/2, \sqrt{3}/2)$ for 60°.



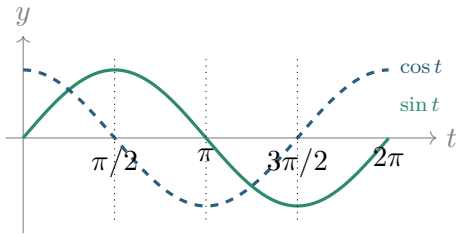
The picture is a reminder that exact values are ratios of lengths: the unit circle merely normalizes the hypotenuse to 1.

1.7 From the circle to a wave

There is one more idea that connects this chapter to the rest of the course. Suppose the angle increases steadily in time:

$$\theta(t) = \omega t, \qquad x(t) = \cos(\omega t), \quad y(t) = \sin(\omega t).$$

The moving point goes around a circle, but the horizontal and vertical coordinates each rise and fall repeatedly. Plotting either coordinate against time produces a wave.



This is a picture of a *coordinate varying over time*, not the circular path itself. The distinction will matter when we study graphs of periodic functions in Module 3.

1.8 A complete mathematical model

A Ferris wheel has radius 8 meters, its center is 10 meters above ground, and a passenger starts at the rightmost point. Suppose it rotates counterclockwise once every 40 seconds. Its angular speed is $\omega = 2\pi/40 = \pi/20$ radians per second, and the passenger's position relative to the ground is

$$\boxed{x(t) = 8\cos\left(\frac{\pi t}{20}\right), \qquad y(t) = 10 + 8\sin\left(\frac{\pi t}{20}\right).}$$

At $t = 0$ the passenger is at $(8, 10)$. At $t = 10$ seconds the wheel has turned through $\pi/2$, so the passenger is at $(0, 18)$. At $t = 20$ seconds the passenger is at $(-8, 10)$. The model answers position questions without tracking the entire path by hand.

Interpreting the constants. The radius appears as the amplitude of each coordinate function. The center height appears as a vertical translation. The revolution time appears in the angular speed. Each physical fact has an identifiable place in the equations.

Try it. A different wheel has radius 6 meters, center height 7 meters, and period 30 seconds. Starting from the rightmost point, write its coordinate model and find the passenger's height after 7.5 seconds.

1.9 Exercises

1. Convert $-135°$ and $300°$ to radians.
2. Convert $11\pi/6$ and $-5\pi/4$ to degrees.
3. Find an angle in $[0, 2\pi)$ coterminal with $19\pi/6$.
4. A circle of radius 8 cm subtends an angle of $5\pi/12$. Find the arc length.
5. A turntable makes $33\frac{1}{3}$ revolutions per minute. Find angular speed in radians per second.
6. Explain why sine and cosine must lie between -1 and 1.
7. Find unit-circle coordinates at $2\pi/3$, $5\pi/4$, and $11\pi/6$.
8. A Ferris wheel has radius 10 m and center $(0, 12)$. Write a rider's coordinates after turning through θ from the rightmost point.
9. A point begins at $(1, 0)$ and turns through $-7\pi/6$ radians. Determine a coterminal angle in $[0, 2\pi)$ and give the exact point.
10. A wheel completes 9 revolutions in 2 minutes. Determine its angular speed in both radians per second and degrees per second.

11. A rider on a circle of radius 4 m moves through 75°. Find the arc length exactly in terms of π and approximately to two decimals.

12. Describe the coordinates at $3\pi/2$ and explain why a terminal point on an axis does not belong to any open quadrant.

13. For the Ferris-wheel model in the chapter, explain which physical quantity would change if the coefficient 8 in both coordinate equations became 12.

Looking ahead. Next week we extend sine and cosine to the remaining four trigonometric functions.

2 The Six Trigonometric Functions

Central question. How do six functions arise naturally from the same moving point on a circle?

The six functions are easier to remember when we begin with one moving point rather than six separate formulas. Picture a point going around the unit circle. Its horizontal coordinate is $x = \cos\theta$ and its vertical coordinate is $y = \sin\theta$. The remaining functions compare those coordinates or take their reciprocals. Thus a single geometric position contains all six pieces of information.

Learning goals. By the end of this chapter, you should be able to define all six functions on the unit circle, find exact values in any quadrant, explain where each function is undefined, sketch their traditional graphs, and use trigonometry to interpret measured lengths and angles.

2.1 From a point to six functions

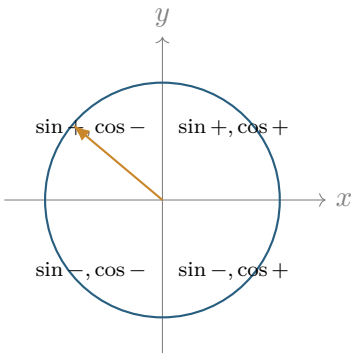
Last week we studied the unit-circle point $P = (x, y) = (\cos\theta, \sin\theta)$. The two coordinates determine four additional ratios whenever the denominators are nonzero:

$$\sin\theta = y, \quad \cos\theta = x, \quad \tan\theta = \frac{y}{x},$$
$$\csc\theta = \frac{1}{y}, \quad \sec\theta = \frac{1}{x}, \quad \cot\theta = \frac{x}{y}.$$

These are not six unrelated measurements. They are six different ways of reading the same geometric information.

Definition 2.1. The *tangent* function is $\tan\theta = \sin\theta/\cos\theta$ where $\cos\theta \neq 0$. The functions csc, sec, cot are the reciprocals of sin, cos, tan respectively, wherever defined.

Example 2.1. At $\theta = \pi/3$ the circle point is $(1/2, \sqrt{3}/2)$. Thus $\tan\theta = \sqrt{3}$, $\sec\theta = 2$, $\csc\theta = 2/\sqrt{3} = 2\sqrt{3}/3$, and $\cot\theta = 1/\sqrt{3} = \sqrt{3}/3$.

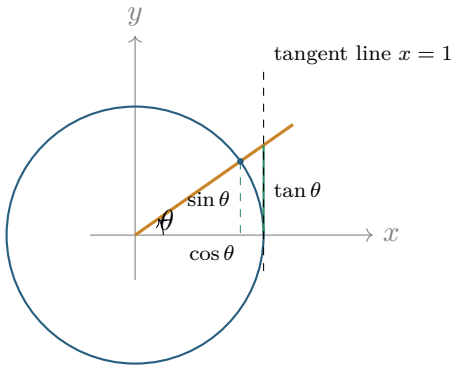


2.2 Why tangent is a slope

Imagine the ray from the origin through the point $P = (\cos\theta, \sin\theta)$. Where cosine is nonzero, the slope of that ray is

$$m = \frac{\text{rise}}{\text{run}} = \frac{\sin\theta}{\cos\theta} = \tan\theta.$$

This connects a familiar idea from linear equations to circular geometry: tangent records the *directional slope* of the ray. A nearly vertical ray has a very large positive or negative slope. A vertical ray has no finite slope at all.



On the vertical line $x = 1$, the extension of the ray reaches height $\tan\theta$. This construction shows why the function is named *tangent*: it can be read along a line tangent to the unit circle.

Example 2.2. At $30°$, the unit-circle coordinates are $(\sqrt{3}/2, 1/2)$. The slope of the corresponding ray is

$$\tan 30° = \frac{1/2}{\sqrt{3}/2} = \frac{\sqrt{3}}{3}.$$

The same slope describes a ramp rising one unit for every $\sqrt{3}$ units of horizontal travel.

2.3 Signs and the quadrants

The coordinate signs determine every trig-function sign. In quadrant I both coordinates are positive; in II, only sine is positive; in III, tangent is positive because both coordinates are negative; and in IV, only cosine is positive. Reciprocal functions have the same signs as their corresponding original functions.

Example 2.3. For $\theta = 5\pi/4$, the reference angle is $\pi/4$ and $P = (-\sqrt{2}/2, -\sqrt{2}/2)$. Therefore $\tan\theta = 1$, $\sec\theta = -\sqrt{2}$, and $\csc\theta = -\sqrt{2}$.

For a useful sign check, consider $150° = 5\pi/6$. The point is $(-\sqrt{3}/2, 1/2)$. Both sine and cosecant are positive, both cosine and secant are negative, and tangent and cotangent are negative:

$$\tan(5\pi/6) = -\frac{\sqrt{3}}{3}, \qquad \sec(5\pi/6) = -\frac{2\sqrt{3}}{3}, \qquad \csc(5\pi/6) = 2.$$

Notice that no signs were memorized separately. All of them followed from the two coordinates.

	sin	cos	tan	csc	sec	cot
Quadrant I	+	+	+	+	+	+
Quadrant II	+	−	−	+	−	−
Quadrant III	−	−	+	−	−	+
Quadrant IV	−	+	−	−	+	−

Try it. The terminal point of an angle lies in quadrant IV and satisfies $\cos\theta = 3/5$. Determine $\sin\theta$, $\tan\theta$, and their reciprocals, showing how the quadrant tells you the signs.

2.4 Where are the functions undefined?

Division by zero has no meaning. In particular, $\tan\theta$ and $\sec\theta$ do not exist where $\cos\theta = 0$, at $\pi/2 + k\pi$. Similarly $\cot\theta$ and $\csc\theta$ do not exist where $\sin\theta = 0$, at $k\pi$, for any integer k. The word *undefined* matters: an undefined value is not the number zero.

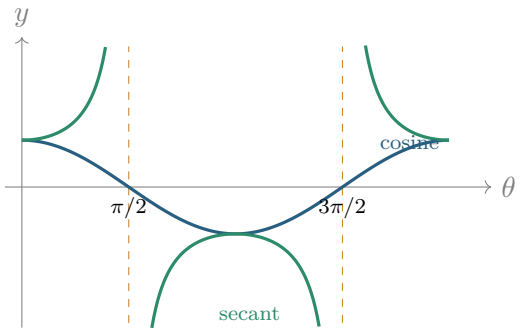
Try it. Find all six trig-function values at $\theta = 3\pi/2$, identifying every undefined value.

2.5 Zeros, poles, and disconnected graphs

A denominator becomes small as its reciprocal becomes large. This is the main visual idea behind tangent, cotangent, secant, and cosecant.

For example, as θ approaches $\pi/2$ from the left, $\cos\theta$ approaches zero from above. Consequently $\sec\theta = 1/\cos\theta$ increases without bound. From the right, cosine is slightly negative, and secant becomes a large negative number. At $\pi/2$ itself, secant has no value.

On an ordinary graph, the vertical line $x = \pi/2$ is a *vertical asymptote* of secant and tangent. The picture shows why a graph cannot be joined through that line.



Here the green curve has been drawn only where its values fit in the viewing window. It continues upward or downward without touching its asymptotes. The graph does *not* cross the asymptote, and we must not connect the separated branches.

Try it. Describe the same phenomenon for $\csc\theta$ near $\theta = 0$. Explain why a zero of sine becomes an undefined value of cosecant.

2.6 Right triangles are a special case

For an acute angle θ in a right triangle, scale the triangle so its hypotenuse has length 1. The opposite and adjacent sides then give the coordinates $\sin\theta$ and $\cos\theta$. Restoring a general hypotenuse h gives

$$\sin\theta = \frac{\text{opposite}}{h}, \quad \cos\theta = \frac{\text{adjacent}}{h}, \quad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

The unit circle is more general than the right-triangle definitions: it handles negative angles, obtuse angles, and angles exceeding a full revolution.

Example 2.4. A ramp rises 1.5 meters over a horizontal run of 6 meters. Its angle of inclination satisfies $\tan\theta = 1.5/6 = 1/4$. Therefore $\theta = \arctan(1/4) \approx 14.0°$.

This gives us the complete right-triangle set of six ratios:

	opposite and hypotenuse	adjacent and hypotenuse	opposite and adjacent
direct	$\sin\theta = o/h$	$\cos\theta = a/h$	$\tan\theta = o/a$
reciprocal	$\csc\theta = h/o$	$\sec\theta = h/a$	$\cot\theta = a/o$

All six ratios come from the same three lengths. The unit-circle definitions extend them beyond acute angles without changing the relationships.

Example 2.5. A support cable stretches 13 meters from the ground to a point 12 meters above it. The horizontal run is $\sqrt{13^2 - 12^2} = 5$ meters. If θ is the angle with the ground, then

$$\sin\theta = \frac{12}{13}, \quad \cos\theta = \frac{5}{13}, \quad \tan\theta = \frac{12}{5}, \quad \csc\theta = \frac{13}{12}, \quad \sec\theta = \frac{13}{5}, \quad \cot\theta = \frac{5}{12}.$$

The reciprocal values describe the same geometry, not new triangles.

2.7 Special angles without memorizing a disconnected table

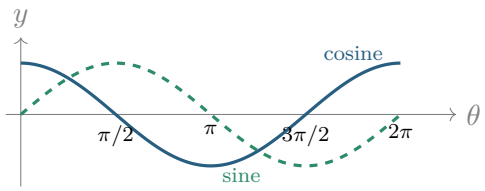
An equilateral triangle of side 2 splits into two right triangles with sides $1, \sqrt{3}, 2$. This produces $\sin 30° = 1/2$ and $\cos 30° = \sqrt{3}/2$. A square of side 1 has diagonal $\sqrt{2}$, producing $\sin 45° = \cos 45° = \sqrt{2}/2$.



2.8 Reading the traditional trigonometric graphs

A graph changes the question we ask. On the unit circle, we hold an angle fixed and locate a point. On a Cartesian graph, we let the angle vary along the horizontal axis and watch one numerical output rise and fall. The graph of a trigonometric function is not the circular path of the point; it is a record of one coordinate or ratio.

Sine and cosine remain between -1 and 1. Both repeat after a full rotation, so their fundamental period is 2π radians, or 360°. Tangent and cotangent repeat after half a turn, so their fundamental period is π radians, or 180°. Secant and cosecant share the 2π period of their reciprocal partners.



Three questions to ask of any trig graph. Where does the function cross the horizontal axis? Where is it undefined? How far must the input angle increase before the graph repeats? Together, these questions reveal most of its global shape.

Example 2.6. Why does tangent repeat after only π? Rotating the unit-circle point by 180° changes both coordinate signs:

$$\frac{\sin(\theta + \pi)}{\cos(\theta + \pi)} = \frac{-\sin\theta}{-\cos\theta} = \tan\theta.$$

The quotient repeats even though the point moves to the opposite side of the circle.

Explore interactively. The course's Six Trigonometric Graphs Explorer lets you compare all six functions. The Reciprocal Trigonometric Graphs Explorer places each function beside its reciprocal; watch how the zeros of one function become asymptotes of the other.

Try it. Sketch $y = \tan x$ and $y = \cot x$ on $(0, \pi)$, marking their zeros and asymptotes. How do the two plots differ at the interval's endpoints?

2.9 Applications: orientation and measurement

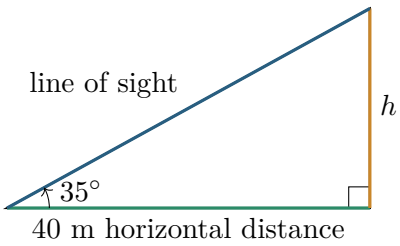
Suppose an observer stands 40 m from the foot of a vertical tower. The angle of elevation to its top is 35°. The tower's height h satisfies

$$\tan 35^\circ = \frac{h}{40}, \qquad h = 40 \tan 35^\circ \approx 28.0 \text{ m}.$$

Here the tangent function links an angle to a ratio of lengths. Always sketch the physical situation and label which distances are opposite and adjacent before calculating.

2.10 Measuring height and choosing the right ratio

In a physical measurement problem, the right triangle should come *before* the calculator. A horizontal distance and a vertical height are related by tangent. A direct line-of-sight distance and a vertical height are related by sine. Recognizing the geometry is more important than memorizing which button to press.



Worked example: estimate a tree height. An observer is 20 meters from the base of a tree and measures an angle of elevation of 40° to its top. If the observer's eyes are 1.6 meters above ground, the elevation of the treetop above eye level is

$$h_{\text{above eyes}} = 20 \tan 40^\circ \approx 16.8 \text{ m}.$$

The tree height is therefore about $16.8 + 1.6 = 18.4$ meters. The final addition matters: a correct trig ratio still needs an accurate interpretation of the measured height.

Try it. A rescue drone is observed at an elevation angle of 25° from an observer whose eye level is 1.7 meters above ground. The point directly beneath the drone is 90 meters away horizontally. Estimate the drone's altitude.

2.11 Exercises

1. At $\theta = 2\pi/3$, find all six functions exactly.
2. At $\theta = 7\pi/4$, find $\tan, \cot, \sec, \csc$.
3. Explain geometrically why $\tan(\pi/2)$ is undefined.

4. For which angles in $[0, 2\pi)$ is $\csc\theta$ undefined?

5. A right triangle has opposite side 5 and adjacent side 12. Find all six trig ratios exactly.

6. A ladder leans against a wall, reaching 8 m high. Its foot is 3 m from the wall. Find the angle with the ground.

7. A drone is viewed at an elevation angle of 28° from a point 75 m horizontally from the point below it. Estimate its altitude.

8. Show that $\sin^2\theta + \cos^2\theta = 1$ implies $1 + \tan^2\theta = \sec^2\theta$ wherever $\cos\theta \neq 0$.

9. Find the exact values of all six trig functions at $11\pi/6$.

10. Describe the zeros and vertical asymptotes of $y = \tan x$ on $[-2\pi, 2\pi]$.

11. Sketch one period of $y = \sec x$, labeling asymptotes and its lowest positive and highest negative values.

12. Explain why neither $\csc x$ nor $\sec x$ can equal $1/2$.

13. If $\cos\theta = -12/13$ and θ lies in quadrant II, determine all other trig values exactly.

14. Show using the unit circle that $\tan(\theta + \pi) = \tan\theta$ wherever both sides are defined.

15. A point on a rotating wheel has unit-circle coordinates $(0, -1)$. Identify the angles in $[0, 2\pi)$ and all four undefined or zero-valued functions.

3 Periodic Functions and Their Graphs

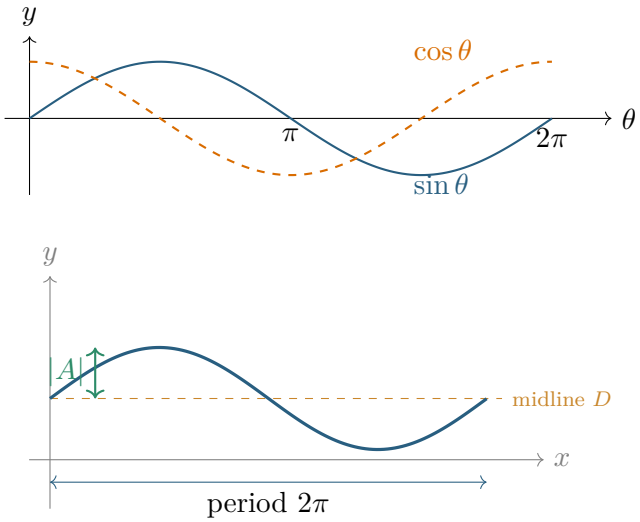
Central question. How does rotation become a repeating signal, and how can we recover the motion from a graph?

3.1 The circle makes a wave

For a point traveling around the unit circle with angle θ, its vertical coordinate is $y = \sin\theta$ and its horizontal coordinate is $x = \cos\theta$. Each revolution restores the same coordinates. Hence

$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

The *period* is the smallest positive input change that repeats the full function. Both sine and cosine have period 2π. Sine starts at 0 and initially rises; cosine starts at 1.



3.2 Amplitude and vertical translation

Consider $y = A\sin x + D$. Its midline is $y = D$, its amplitude is $|A|$, and its range is $[D - |A|, D + |A|]$. A negative A reflects the wave across the midline.

Example 3.1. The function $y = 3\sin x - 2$ has amplitude 3, midline $y = -2$, maximum 1, and minimum -5.

3.3 Period and horizontal scaling

For $y = \sin(Bx)$ with $B \neq 0$, one full cycle occurs when Bx changes by 2π; the period is $T = 2\pi/|B|$.

Example 3.2. The graph of $y = \cos(4x)$ repeats every $\pi/2$ units of input.

When x denotes time measured in seconds, the quantity $|B|$ is angular frequency in radians per second, and ordinary frequency is $f = 1/T = |B|/(2\pi)$ cycles per second.

3.4 Phase shift

A useful general form is

$$\boxed{y = A\sin(B(x - C)) + D}.$$

Here C is the horizontal shift, D the midline, $|A|$ the amplitude, and $2\pi/|B|$ the period. We should distinguish $B(x - C)$ from $Bx - C$: factoring first prevents an incorrect phase shift.

Example 3.3. For $y = 2\sin(3x - \pi) + 1 = 2\sin(3(x - \pi/3)) + 1$, the period is $2\pi/3$, the shift is $\pi/3$ to the right, the amplitude is 2, and the midline is 1.

3.5 Five landmark points

Rather than sketching a sinusoid by guesswork, divide one period into four equal intervals. For $A > 0$, sine begins on the midline, rises to a maximum after one quarter-period, returns to the midline, reaches a minimum, then returns to the midline. A negative amplitude reverses the rises and falls.

Try it. Find five landmark points of $y = -2\cos(\pi(x - 1)) + 3$, showing one entire period.

3.6 Models of repeating motion

A Ferris wheel has radius 9 m, its center is 11 m above ground, and it makes one revolution every 30 s. If a passenger begins at the rightmost position and travels counterclockwise, their height is

$$h(t) = 11 + 9\sin\left(\frac{2\pi}{30}t\right).$$

Its amplitude is 9 m, midline is 11 m, and period is 30 s. A sinusoidal model can describe a wheel, a spring, a tide, or a musical tone, provided its parameters actually correspond to the chosen physical quantities.

3.7 Tangent and reciprocal graphs

The tangent function repeats after π, not 2π, because both sine and cosine change sign after a half turn and their ratio remains the same:

$$\tan(x + \pi) = \tan x.$$

Tangent has vertical asymptotes at $x = \pi/2 + k\pi$. The secant and cosecant graphs arise as reciprocals of cosine and sine; their breaks occur wherever the respective denominator is zero.

3.8 Exercises

1. Find the amplitude, midline, period, and range of $y = -4\sin(2x) + 1$.
2. Rewrite $y = 3\cos(5x - \pi) + 2$ in shifted form and identify the phase shift.
3. Give five landmark points for $y = 2\sin(x - \pi/2)$.
4. Explain why $y = \sin(2x)$ completes two periods on $[0, 2\pi]$.
5. A wheel of radius 6 m rotates once every 24 seconds. Its center is 8 m above the ground. Model the rider's height when starting at the rightmost point.
6. Identify the vertical asymptotes of $y = \tan(2x)$ on $[0, 2\pi)$.
7. Sketch one period of $y = 2\sec x$ and indicate all discontinuities.
8. Explain how you can determine amplitude and midline directly from observed maximum and minimum values.

4 Inverse Trigonometry and Identities

Central question: How can we recover an angle from its trigonometric value, and how do we prove two expressions represent the same function?

4.1 An inverse is not a reciprocal

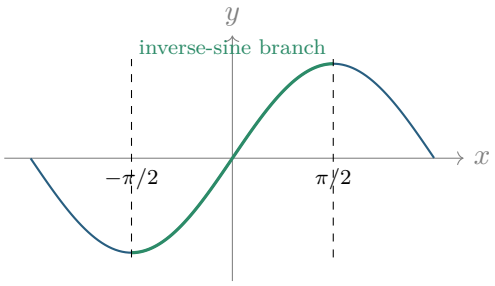
The notation $\sin^{-1} x$ means the inverse function $\arcsin x$. It does not mean $1/\sin x = \csc x$. The inverse answers an angle question; the reciprocal answers a ratio question.

Sine is periodic, so an inverse function requires choosing an interval on which sine is one-to-one. We use $[-\pi/2, \pi/2]$. In exactly the same way, cosine uses $[0, \pi]$ and tangent uses $(-\pi/2, \pi/2)$.

$f(x)$	Domain	Range
$\arcsin x$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\arccos x$	$[-1, 1]$	$[0, \pi]$
$\arctan x$	$\mathbb{R}$	$(-\pi/2, \pi/2)$

Example 4.1. $\arcsin(1/2) = \pi/6$, $\arccos(-1/2) = 2\pi/3$, and $\arctan(-1) = -\pi/4$.

Example 4.2. $\arcsin(\sin(5\pi/6)) = \pi/6$, not $5\pi/6$, because the inverse reports its principal value.



4.2 Compositions and triangle geometry

The identity $\sin(\arcsin x) = x$ holds for $-1 \le x \le 1$, but $\arcsin(\sin\theta) = \theta$ only for θ in the principal range. Suppose $\theta = \arctan(3/4)$. Then a right triangle has legs $3, 4$ and hypotenuse 5, so

$$\sin(\arctan(3/4)) = \frac{3}{5}.$$

4.3 An identity is true throughout its domain

From the unit-circle equation we obtain

$$\boxed{\sin^2 x + \cos^2 x = 1}.$$

Dividing by $\cos^2 x$ (only where $\cos x \neq 0$) gives $1 + \tan^2 x = \sec^2 x$. Dividing by $\sin^2 x$ gives $1 + \cot^2 x = \csc^2 x$ where $\sin x \neq 0$.

Example 4.3. For $\sin x \neq 0$,

$$\frac{1 - \cos^2 x}{\sin x} = \frac{\sin^2 x}{\sin x} = \sin x.$$

This is an algebraic proof on the shared domain, not a numerical check at a few inputs.

4.4 Angle addition and double angles

Two successive plane rotations produce the sum identities

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

Set $\beta = -\gamma$ for difference identities and $\beta = \alpha$ for double-angle identities:

$$\sin(2x) = 2\sin x\cos x, \qquad \cos(2x) = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x.$$

Example 4.4. $\sin(75^\circ) = \sin(45^\circ + 30^\circ) = (\sqrt{6}+\sqrt{2})/4$.

4.5 Applications: measured angles

An observation of a tower from 40 m away gives $\tan\theta = 3/4$. The elevation angle is $\arctan(3/4) \approx 36.9^\circ$, while the height is exactly $40(3/4) = 30$ m. Inverse functions recover angles; ratios can often recover distances directly.

4.6 Exercises

1. Evaluate exactly $\arcsin(-\sqrt{2}/2)$ and $\arctan(\sqrt{3})$.
2. Explain why $\arcsin(\sin(3\pi/4)) \neq 3\pi/4$.
3. Find $\cos(\arcsin(5/13))$ with a triangle.
4. Find $\sin(\arctan(-2))$ exactly, including its sign.
5. Prove $(\sec^2 x - 1)/\tan x = \tan x$ on its domain.
6. Prove $(\sin x + \cos x)^2 = 1 + \sin(2x)$.
7. Calculate $\cos(75^\circ)$ exactly by angle addition.
8. Express $\cos(2x)$ using $\sin x$ alone.

5 Trigonometric Equations and Applications

Central question. When a trigonometric function has a prescribed value, how can we find *all* the angles that produce it? How can we use angles to solve triangles that are not right triangles?

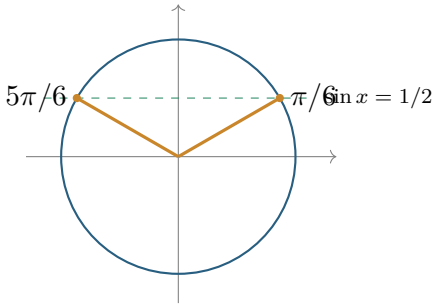
5.1 A first equation on the unit circle

Consider $\sin x = 1/2$. The unit circle has two points whose vertical coordinate is $1/2$, at $x = \pi/6$ and $x = 5\pi/6$. Every full revolution reproduces these points:

$$\boxed{x = \frac{\pi}{6} + 2\pi k \quad \text{or} \quad x = \frac{5\pi}{6} + 2\pi k, \qquad k \in \mathbb{Z}.}$$

An inverse trigonometric function gives one principal angle, not necessarily every solution. The unit circle and periodicity give the complete answer.

Example 5.1. Solve $\cos x = -\sqrt{2}/2$ for $0 \le x < 2\pi$. The solutions are $3\pi/4$ and $5\pi/4$.



5.2 Equations with transformations

To solve $\sin(2x) = \sqrt{3}/2$ on $[0, 2\pi)$, first solve for $u = 2x$ on $[0, 4\pi)$. We obtain $u = \pi/3, 2\pi/3, 7\pi/3, 8\pi/3$ and therefore

$$x = \frac{\pi}{6}, \frac{\pi}{3}, \frac{7\pi}{6}, \frac{4\pi}{3}.$$

A changed frequency changes how many full cycles fit into the requested interval.

Example 5.2. Solve $2\cos x - 1 = 0$ on $[0, 2\pi)$. Isolating cosine gives $\cos x = 1/2$, hence $x = \pi/3$ or $5\pi/3$.

5.3 Factoring and checking domains

An equation may contain more than one trig function. Algebraic techniques still work:

$$2\sin^2 x - \sin x = 0 \quad \Longrightarrow \quad \sin x(2\sin x - 1) = 0.$$

On $[0, 2\pi)$ we obtain $x = 0, \pi, \pi/6, 5\pi/6$. Dividing by $\sin x$ at the outset would have lost two solutions.

5.4 Trigonometric identities in equations

For example, $2\cos^2 x - 1 = 0$ becomes $\cos^2 x = 1/2$, hence $\cos x = \pm\sqrt{2}/2$. There are four solutions on $[0, 2\pi)$. Using identities can make equations simpler, but each transformation should preserve the original domain.

5.5 Solving triangles: notation

Write the vertices of a triangle as A, B, C, and the opposite side lengths as a, b, c. Its angles satisfy $A + B + C = \pi$. For right triangles, the six trig ratios are enough. For oblique triangles, two laws extend our tools.

Definition 5.1 (Law of Sines)**.** For every nondegenerate triangle,

$$\boxed{\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.}$$

Example 5.3. A triangle has $A = 40°$, $B = 65°$, and $a = 9$ cm. Since $C = 75°$, the Law of Sines gives $b = 9 \sin 65°/\sin 40° \approx 12.69$ cm.

5.6 The ambiguous SSA case

Given two sides and a nonincluded angle (SSA), the Law of Sines may produce zero, one, or two triangles. If $\sin B = s$, both $B = \arcsin s$ and $B = \pi - \arcsin s$ must be considered, and each candidate must fit with the known angle A in a triangle.

Example 5.4. Suppose $A = 30°$, $a = 5$ and $b = 8$. Then $\sin B = 8 \sin 30°/5 = 0.8$. Both $B \approx 53.13°$ and $B \approx 126.87°$ are consistent with $A + B < 180°$, so two distinct triangles exist.

Definition 5.2 (Law of Cosines)**.** For a triangle with sides a, b, c opposite angles A, B, C,

$$\boxed{c^2 = a^2 + b^2 - 2ab \cos C.}$$

When $C = 90°$ this becomes the Pythagorean theorem. The cosine term measures how the included angle modifies the length of the opposite side.

Example 5.5. Two sides of a triangle are 7 m and 10 m, with included angle $60°$. The opposite side satisfies $c^2 = 49 + 100 - 140(1/2) = 79$, so $c = \sqrt{79}$ m.

5.7 Applied modeling and interpretation

Two sensors observe a distant object from different locations. The baseline and their sight angles make a triangle. The Law of Sines can recover an inaccessible distance if we know a side and two angles. Always include the diagram, angle units, and a reasonableness check.

5.8 Exercises

1. Solve $\sin x = -\sqrt{3}/2$ for $0 \le x < 2\pi$.
2. Solve $2\sin(3x) = 1$ for $0 \le x < 2\pi$; find every solution.
3. Solve $2\cos^2 x - \cos x = 0$ on $[0, 2\pi)$.
4. Solve $\tan x = -1$ for all real x.
5. A triangle has $A = 35°$, $B = 80°$ and $a = 12$. Find C and b.
6. For $A = 30°$, $a = 5$, $b = 8$, calculate both possible values of C in the SSA case.
7. Find the third side of a triangle when two sides have length 6 and 9 and their included angle is $120°$.

8. Explain why the Law of Cosines becomes the Pythagorean theorem for right triangles.

6 Vectors and Geometric Applications

Central question. How do we describe a quantity that has both a size and a direction?

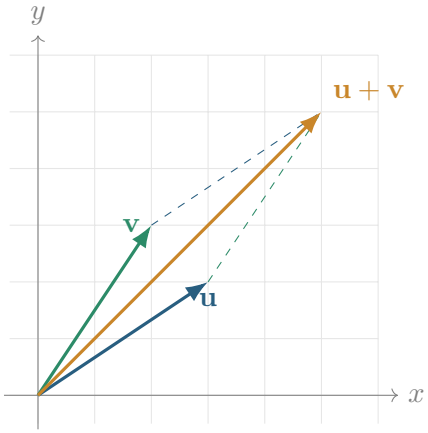
6.1 From coordinates to directed displacements

In the first half of the course we used $(\cos\theta, \sin\theta)$ to locate a point on a circle. A vector uses the same coordinates to describe a *displacement*: how far to move horizontally and vertically. Write $\mathbf{v} = \langle a, b\rangle$. Its magnitude is

$$\boxed{\|\mathbf{v}\| = \sqrt{a^2 + b^2}}.$$

Vectors with the same components represent the same displacement even when drawn at different starting points.

Example 6.1. A walker moves 3 km east and 4 km north. The resulting vector is $\langle 3, 4\rangle$ km and its magnitude is 5 km. The route traveled is 7 km, while the displacement is only 5 km.



6.2 Vector arithmetic

Addition combines displacements:

$$\langle a, b\rangle + \langle c, d\rangle = \langle a + c, b + d\rangle.$$

Scalar multiplication changes their length, and a negative scalar reverses direction:

$$k\langle a, b\rangle = \langle ka, kb\rangle.$$

Subtracting vectors compares locations. If $P = (p_1, p_2)$ and $Q = (q_1, q_2)$, then $\overrightarrow{PQ} = \langle q_1 - p_1, q_2 - p_2\rangle$.

Example 6.2. For $\mathbf{u} = \langle 2, -1\rangle$ and $\mathbf{v} = \langle -3, 4\rangle$, we have $\mathbf{u} + \mathbf{v} = \langle -1, 3\rangle$ and $2\mathbf{u} - \mathbf{v} = \langle 7, -6\rangle$.

6.3 Direction and trigonometry

A unit vector has magnitude 1. The unit vector pointing at angle θ is

$$\mathbf{e}_\theta = \langle \cos\theta, \sin\theta\rangle.$$

A vector of magnitude r in that direction is

$$\boxed{\mathbf{v} = r\langle \cos\theta, \sin\theta\rangle}.$$

Conversely, for a nonzero vector $\langle a, b\rangle$, the angle must be chosen in the appropriate quadrant. An inverse tangent applied only to b/a may miss that quadrant.

Example 6.3. A force of 20 N directed 30° above the horizontal has components $\langle 10\sqrt{3}, 10\rangle$ N.

6.4 The dot product measures alignment

Define

$$\mathbf{u}\cdot\mathbf{v} = u_1v_1 + u_2v_2.$$

The geometric interpretation is

$$\boxed{\mathbf{u}\cdot\mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\|\cos\phi},$$

where ϕ is the angle between nonzero vectors. Positive dot product means an acute angle; negative means an obtuse angle; zero means perpendicular vectors.

Example 6.4. With $\mathbf{u} = \langle 3, 4\rangle$ and $\mathbf{v} = \langle 4, -3\rangle$, the dot product is $12 - 12 = 0$. Both vectors are nonzero, so they are perpendicular.

6.5 Projection and useful components

The scalar component of $\mathbf{u}$ in the direction of a nonzero vector $\mathbf{v}$ is

$$\operatorname{comp}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|}.$$

The corresponding vector projection is

$$\operatorname{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|^2}\mathbf{v}.$$

Projection is the part of one vector aligned with another. In physics, the work done by a constant force $\mathbf{F}$ across displacement $\mathbf{d}$ is $W = \mathbf{F}\cdot\mathbf{d}$, measured in joules when force is in newtons and displacement in meters.

Example 6.5. A force $\langle 10, 5\rangle$ N acts during displacement $\langle 3, 0\rangle$ m. The work is 30 J: only the horizontal force component contributes.

6.6 A geometric connection to the Law of Cosines

For $\mathbf{u}, \mathbf{v}$ we have

$$\|\mathbf{u}-\mathbf{v}\|^2 = (\mathbf{u}-\mathbf{v})\cdot(\mathbf{u}-\mathbf{v}) = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\|\cos\phi.$$

Thus the Law of Cosines from Module 5 is also a theorem about vector differences.

6.7 Exercises

1. Find the vector from $P = (-2, 3)$ to $Q = (5, -1)$ and its magnitude.
2. For $\mathbf{u} = \langle 2, 5\rangle$, $\mathbf{v} = \langle -1, 3\rangle$, compute $\mathbf{u} + \mathbf{v}$ and $3\mathbf{u} - 2\mathbf{v}$.
3. Find exact components of a displacement of magnitude 12 at angle 150°.
4. Find a unit vector in the direction of $\langle -3, 4\rangle$.
5. Compute $\langle 2, -1\rangle \cdot \langle 5, 3\rangle$ and interpret its sign.
6. Determine whether $\langle 6, 2\rangle$ and $\langle -1, 3\rangle$ are perpendicular.

7. Find $\text{proj}_{\mathbf{v}} \mathbf{u}$ when $\mathbf{u} = \langle 4, 3 \rangle$ and $\mathbf{v} = \langle 1, 0 \rangle$.

8. A force of 50 N acts 60° from the direction of a 4 m displacement. Calculate the work done.

7 Polar Coordinates and Complex Numbers

Central question. What happens when we describe a point by its distance and direction instead of its horizontal and vertical coordinates?

7.1 Two descriptions of the same point

Cartesian coordinates specify (x, y). Polar coordinates specify (r, θ), where r is the directed radial distance and θ is an angle from the positive horizontal axis:

$$\boxed{x = r\cos\theta, \qquad y = r\sin\theta.}$$

For $r \geq 0$, the distance is $r = \sqrt{x^2 + y^2}$. The angle must be chosen using the signs of both coordinates, not just $\arctan(y/x)$.

Example 7.1. The polar point $(4, 5\pi/6)$ corresponds to $(x, y) = (-2\sqrt{3}, 2)$. Its distance from the origin is 4, and it lies in Quadrant II.

Try it. Convert $(r, \theta) = (6, -\pi/3)$ to Cartesian coordinates. Then describe its direction using an angle in $[0, 2\pi)$.

7.2 Polar coordinates are not unique

Angles that differ by 2π represent the same direction, so (r, θ) and $(r, \theta + 2\pi k)$ describe the same point. A negative radius reverses the ray:

$$(r, \theta) = (-r, \theta + \pi).$$

For example, $(2, \pi/3)$, $(2, 7\pi/3)$, and $(-2, 4\pi/3)$ are three descriptions of the same point. At the origin $r = 0$, every angle describes the same location.

7.3 Complex numbers as plane vectors

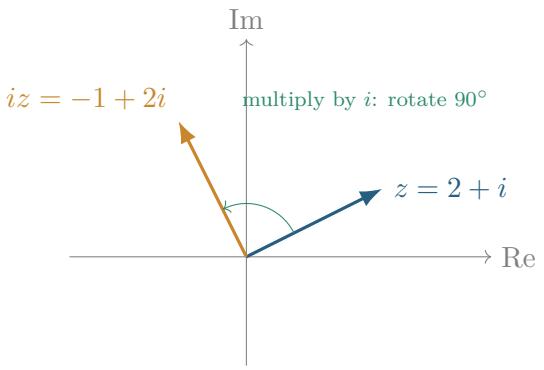
Introduce i satisfying $i^2 = -1$. A complex number $z = a + bi$ corresponds to the point (a, b) in the complex plane. Its modulus and argument are

$$|z| = \sqrt{a^2 + b^2}, \qquad \arg z = \theta \quad \text{when } z = |z|(\cos\theta + i\sin\theta).$$

The argument of $z = 0$ is undefined. As with polar angles, nonzero arguments are only unique after choosing a branch, such as $(-\pi, \pi]$.

Example 7.2. For $z = -\sqrt{3} + i$, the modulus is 2 and one argument is $5\pi/6$. Thus

$$z = 2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right).$$



7.4 Addition and multiplication mean different geometries

Complex addition follows the vector rule:

$$(a+bi)+(c+di)=(a+c)+(b+d)i.$$

Complex multiplication uses $i^2=-1$:

$$(a+bi)(c+di)=(ac-bd)+(ad+bc)i.$$

If $z=r(\cos\alpha+i\sin\alpha)$ and $w=s(\cos\beta+i\sin\beta)$, the angle-addition formulas give

$$\boxed{zw=rs(\cos(\alpha+\beta)+i\sin(\alpha+\beta)).}$$

Therefore multiplying complex numbers multiplies distances from the origin and *adds angles*. A unit complex number acts as a rotation. This unifies the unit circle from Module 1 with vector geometry from Module 6.

Example 7.3. Multiplying $2+i$ by i gives $i(2+i)=-1+2i$. The vector $(2,1)$ rotates counterclockwise by 90° to $(-1,2)$, without changing its length.

7.5 De Moivre's theorem and roots

For integer n, repeated multiplication gives

$$\boxed{[r(\cos\theta+i\sin\theta)]^n=r^n(\cos n\theta+i\sin n\theta)}$$

whenever the expression makes sense; for negative n, require $r>0$. For example, $(\cos(\pi/6)+i\sin(\pi/6))^3=i$. To solve $z^n=R(\cos\phi+i\sin\phi)$ with $R>0$, take

$$z_k=R^{1/n}\left(\cos\frac{\phi+2\pi k}{n}+i\sin\frac{\phi+2\pi k}{n}\right),\qquad k=0,\dots,n-1.$$

These n distinct roots are equally spaced around a circle.

Example 7.4. The four fourth roots of 1 are $1,i,-1,-i$. They are separated by a quarter-turn.

7.6 Polar graphs: a different way to draw a curve

In a polar equation such as $r=2\cos\theta$, the radius changes with the angle. Multiply by r: $r^2=2r\cos\theta$, or $x^2+y^2=2x$. Completing the square yields

$$(x-1)^2+y^2=1.$$

Thus a simple polar equation can represent an ordinary circle displaced from the origin.

7.7 Exercises

1. Convert $(5,2\pi/3)$ to Cartesian coordinates.
2. Give two distinct polar coordinate representations of $(\sqrt{2},\sqrt{2})$.
3. Write $-1-\sqrt{3}i$ in polar form with argument in $[0,2\pi)$.
4. Calculate $(2+i)(1-3i)$ in rectangular form.

5. Interpret multiplication by $-i$ as a plane transformation.
6. Evaluate $(\cos(\pi/4) + i\sin(\pi/4))^4$.
7. Find all three cube roots of 1 and plot them on the unit circle.
8. Convert the polar curve $r = 4\sin\theta$ into a Cartesian equation and identify the curve.

8 Parametric Equations and Conic Sections

Central question. How can the same curve be described by coordinates, by a moving point, or by a geometric rule?

8.1 A moving point traces a curve

In a parametric description, both coordinates depend on a parameter t:

$$\boxed{x = f(t), \qquad y = g(t).}$$

Think of t as time when a particle moves. The curve records its location, while the parameterization also records direction and timing.

Example 8.1. For $x = 2t + 1$, $y = t - 3$, eliminating $t = (x - 1)/2$ gives $y = (x - 7)/2$. If t increases, the point moves up and right along this line.

8.2 Eliminating the parameter

For $x = 3\cos t$, $y = 2\sin t$, square the normalized equations and add:

$$\boxed{\frac{x^2}{9} + \frac{y^2}{4} = 1.}$$

The equation gives the whole ellipse when t ranges over $[0, 2\pi]$. A restricted interval, such as $[0, \pi]$, traces only its upper half. Different parameterizations may have the same image but different speeds or directions.

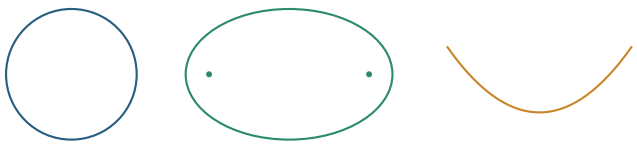
Example 8.2. Both $(x, y) = (\cos t, \sin t)$ and $(x, y) = (\cos 2t, \sin 2t)$ trace the unit circle for $0 \le t \le 2\pi$; the second makes two revolutions.

8.3 Projectile motion as a parametric curve

Under constant downward acceleration g, a projectile launched from (x_0, y_0) at speed v_0 and angle θ satisfies

$$x(t) = x_0 + v_0 \cos\theta\, t, \qquad y(t) = y_0 + v_0 \sin\theta\, t - \tfrac{1}{2} g t^2.$$

When the horizontal velocity is nonzero, eliminating t gives a parabola. The motion model adds information the Cartesian curve alone does not supply, including direction and location at a given time.



circle ellipse with foci parabola

8.4 Conics: circles and ellipses

A *circle* is the set of points at a fixed distance r from a center (h, k):

$$(x-h)^2 + (y-k)^2 = r^2.$$

An *ellipse* can be defined by a constant sum of distances to two foci. In standard form, for $a \geq b > 0$,

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1.$$

Its horizontal semiaxis is a and vertical semiaxis is b; when the major axis is horizontal, focal distance obeys $c^2 = a^2 - b^2$.

Example 8.3. For $x^2/25 + y^2/9 = 1$, the vertices are $(\pm 5, 0)$, and the foci are $(\pm 4, 0)$.

8.5 Parabolas and their focus

A *parabola* is the set of points equidistant from a fixed point (focus) and a fixed line (directrix). The equation

$$(x-h)^2 = 4p(y-k)$$

has vertex (h, k), focus $(h, k+p)$, and directrix $y = k - p$. Its axis is vertical; $p > 0$ opens upward and $p < 0$ downward.

Example 8.4. For $x^2 = 8y$, we have $4p = 8$, so the focus is $(0, 2)$ and the directrix is $y = -2$.

8.6 Hyperbolas and differences of distances

A *hyperbola* consists of points for which the absolute difference of distances to two foci is fixed. A horizontal hyperbola has standard form

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1.$$

Its vertices are $(h \pm a, k)$, its foci are $(h \pm c, k)$ where $c^2 = a^2 + b^2$, and its asymptotes are $y - k = \pm (b/a)(x-h)$.

Example 8.5. The hyperbola $x^2/9 - y^2/16 = 1$ has vertices $(\pm 3, 0)$, foci $(\pm 5, 0)$, and asymptotes $y = \pm 4x/3$.

8.7 Changing descriptions

A parametric curve may be easier to generate than to recognize from a Cartesian equation. A Cartesian equation reveals symmetry or an axis. A focal definition explains why the curve arises in optics, navigation, and geometry. None of these descriptions replaces the others; each answers a different question.

8.8 Exercises

1. Eliminate t from $x = t + 2$, $y = 3t - 1$ and describe the motion as t increases.

2. Find the Cartesian equation traced by $x = 4\cos t$, $y = 3\sin t$.

3. Explain how $(\cos t, \sin t)$ and $(\cos 3t, \sin 3t)$ differ as parameterizations.

4. A projectile leaves the origin at 20 m/s at 30°, with $g = 9.8$ m/s^2. Write its parametric equations.
5. Identify center and radius of $(x-2)^2 + (y+1)^2 = 16$.
6. Find the foci of $x^2/36 + y^2/20 = 1$.
7. Find the focus and directrix of $(x-1)^2 = 12(y+2)$.
8. Find the vertices and asymptotes of $x^2/4 - y^2/9 = 1$.

9 Sequences and Series

Central question. How do repeated steps produce patterns, and how can we predict both a later value and a cumulative total?

9.1 A sequence is an ordered list

A sequence assigns a number a_n to each positive integer n. For example, $a_n = 3 + 2(n-1)$ produces $3, 5, 7, 9, \ldots$. We distinguish the *term* a_n from a *partial sum* $S_n = a_1 + \cdots + a_n$.

9.2 Arithmetic sequences: repeated addition

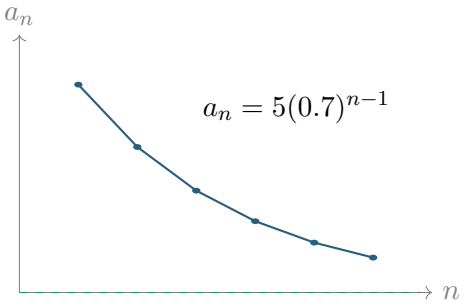
An arithmetic sequence has a fixed common difference d:

$$a_{n+1} = a_n + d, \qquad \boxed{a_n = a_1 + (n-1)d}.$$

Example 9.1. A theater's first row has 12 seats, with two more seats in each successive row. Row 15 has $12 + 14(2) = 40$ seats.

Adding the first and last terms in reverse order gives

$$\boxed{S_n = \frac{n}{2}(a_1 + a_n) = \frac{n}{2}(2a_1 + (n-1)d)}.$$



9.3 Geometric sequences: repeated multiplication

A geometric sequence has a fixed common ratio q:

$$a_{n+1} = qa_n, \qquad \boxed{a_n = a_1 q^{n-1}}.$$

When $q > 1$, positive terms grow exponentially. When $0 < q < 1$, they decay toward zero.

Example 9.2. A population of 500 organisms increases by 8 percent each generation. The model is $a_n = 500(1.08)^{n-1}$ if generation 1 is the initial observation.

For $q \neq 1$, multiply the finite sum by q and subtract:

$$S_n = a_1 + a_1 q + \cdots + a_1 q^{n-1}, \qquad \boxed{S_n = a_1 \frac{1 - q^n}{1 - q}}.$$

If $q = 1$, the sum is na_1. The formulas refer to *finite* sums and work for ratios outside the interval $(0, 1)$ as well.

9.4 Sigma notation

The compact expression $\sum_{k=1}^{n} a_k$ means add the terms for integer indices $k = 1, \dots, n$. For instance,

$$\sum_{k=1}^{4} (2k+1) = 3 + 5 + 7 + 9 = 24.$$

The symbol $\sum$ tells us to add, while its bounds tell us which indices are included.

9.5 Infinite geometric series

If $|q| < 1$, then $q^n \to 0$. The finite geometric sum approaches

$$\boxed{\sum_{n=0}^{\infty} aq^n = \frac{a}{1-q}}, \qquad |q| < 1.$$

The condition matters: the series $1 + 2 + 4 + \cdots$ does not approach a finite sum.

Example 9.3. A ball bounces to 60% of its previous height. Released from height 5 m, the total vertical travel after infinitely many idealized bounces is $5+2(5)(0.6+0.6^2+\cdots) = 5+10(0.6/0.4) = 20$ m.

9.6 Recursive rules beyond arithmetic and geometric patterns

Not every sequence is arithmetic or geometric. Consider

$$a_{n+1} = \frac{a_n + 6}{2}, \qquad a_1 = 0.$$

The first terms are $0, 3, 4.5, 5.25, \dots$. Let $e_n = a_n - 6$ denote the difference from the fixed point. Then $e_{n+1} = e_n/2$, so $e_n = -6(1/2)^{n-1}$ and

$$a_n = 6 - 6(1/2)^{n-1}.$$

The original recursion is not geometric, but its *error from equilibrium* is. This is a useful bridge to feedback and dynamical models.

9.7 Exercises

1. Write the first five terms of $a_n = 7 - 3(n-1)$ and find a_{20}.
2. An arithmetic sequence has $a_1 = 4$, $d = 5$. Calculate S_{30}.
3. A geometric sequence has $a_1 = 81$, $q = 1/3$. Find a_6.
4. Compute $\sum_{k=0}^{5} 2(3/2)^k$ exactly.
5. Explain the difference between $\sum_{k=1}^{n} k$ and its final term n.
6. Determine whether $5 + 5/2 + 5/4 + \cdots$ converges and find its sum.
7. A machine loses 12 percent of its value per year. Express its value after n years if it initially costs \$12,000.
8. For $a_{n+1} = (a_n + 10)/2$, $a_1 = 2$, find the fixed point and an explicit formula for a_n.

10 Synthesis: Motion and Patterns

Central question. How can the different mathematical languages of this course describe one evolving system?

10.1 One motion, several descriptions

Imagine a point traveling counterclockwise around a circle of radius R with angular speed ω. Its angle at time t is $\theta(t) = \theta_0 + \omega t$. In parametric form,

$$x(t) = R\cos(\theta_0 + \omega t), \qquad y(t) = R\sin(\theta_0 + \omega t).$$

Its polar form is $(r, \theta) = (R, \theta_0 + \omega t)$, and its complex form is

$$\boxed{z(t) = R\big(\cos(\theta_0 + \omega t) + i\sin(\theta_0 + \omega t)\big)}.$$

These are not competing models: they encode the same physical position in different coordinate languages.

10.2 Velocity, vectors, and periodicity

Over a short time interval Δt, the displacement vector is

$$\Delta\mathbf{r} = \langle x(t + \Delta t) - x(t), y(t + \Delta t) - y(t)\rangle.$$

For constant angular speed the point completes a revolution after $T = 2\pi/|\omega|$ when $\omega \neq 0$. Its horizontal and vertical coordinates repeat with that period. The direction of motion is tangent to the circle and perpendicular to the radius: this is the geometric heart of circular motion.

10.3 Sampling creates a sequence

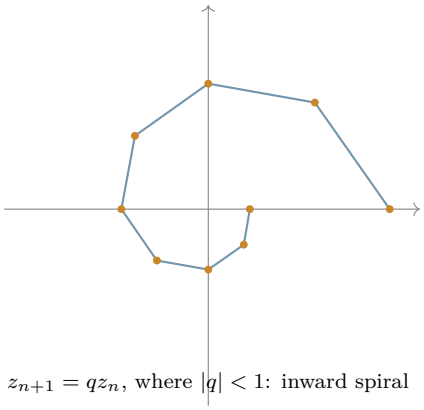
Suppose we record the position every Δt seconds. The resulting sequence of complex positions is

$$z_n = R\big(\cos(\theta_0 + n\delta) + i\sin(\theta_0 + n\delta)\big), \qquad \delta = \omega\Delta t.$$

Writing $u = \cos\delta + i\sin\delta$ gives

$$\boxed{z_{n+1} = uz_n, \qquad z_n = u^n z_0.}$$

De Moivre's theorem then explains every sample as a repeated rotation. If $\delta = 2\pi/m$ for a positive integer m, the sampled orbit closes after m steps (provided this is the smallest such positive step count). If $\delta/(2\pi)$ is irrational, no two different samples have exactly the same position.



$z_{n+1} = qz_n$, where $|q| < 1$: inward spiral

10.4 Scaling and rotation together

Now consider the recurrence $z_{n+1} = qz_n$ where

$$q = \rho(\cos\delta + i\sin\delta).$$

Its explicit solution is $z_n = q^n z_0$. Consequently,

$$|z_n| = \rho^n |z_0|, \qquad \arg z_n \equiv \arg z_0 + n\delta \pmod{2\pi}$$

for nonzero terms. If $0 < \rho < 1$, the points spiral toward the origin; if $\rho > 1$, they spiral outward; and if $\rho = 1$, they remain on a circle. Thus a geometric sequence and a rotation combine to produce a spiral.

10.5 Sums represent accumulated effects

If a system moves by step vectors $\mathbf{v}_0, \mathbf{v}_1, \ldots$, its total displacement after N steps is the vector sum $\sum_{n=0}^{N-1} \mathbf{v}_n$. For a complex-valued geometric pattern $\Delta z_n = aq^n$ with $q \neq 1$,

$$z_N - z_0 = \sum_{n=0}^{N-1} aq^n = a\frac{1-q^N}{1-q}.$$

Notice that the location sequence z_n and the step sequence Δz_n are different objects. Careful modeling starts by distinguishing what is being measured.

10.6 Different curves arise from different constraints

Circular and elliptical motion can be expressed parametrically. A projectile follows a parabola when horizontal velocity is constant and vertical acceleration is constant. Polar formulas express rays and radial patterns directly. The useful representation depends on whether we need a shape, a position, a direction, or a rule for change.

10.7 A complete example

A sensor begins at $z_0 = 2$ in the complex plane. Every second its controller multiplies the position by $q = \frac{1}{2}(\cos 60^\circ + i\sin 60^\circ)$. After n steps,

$$z_n = 2\left(\frac{1}{2}\right)^n (\cos(60^\circ n) + i\sin(60^\circ n)).$$

After three steps the modulus is $1/4$, and the direction is 180°. Hence $z_3 = -1/4$. The modulus obeys a geometric sequence, the direction obeys an arithmetic progression of angles, and the position follows a discrete inward spiral.

10.8 Exercises

1. Write Cartesian, polar, and complex formulas for a circle of radius 5 rotating once every 20 seconds, starting on the positive horizontal axis.

2. Sample that motion every five seconds. Give its first five complex positions and identify its repetition period in samples.

3. For $z_{n+1} = iz_n$ and $z_0 = 1 + i$, calculate $z_1, \ldots, z_4$ and describe the geometry.

4. For $z_{n+1} = (1/2)iz_n$, $z_0 = 4$, find z_4 and explain the change in magnitude and argument.
5. A point travels along $x = 3\cos t$, $y = 2\sin t$. Eliminate the parameter and describe one complete traversal.
6. Explain why a constant-speed circular trajectory is periodic but the spiral $z_n = (0.8e^{i\pi/3})^n z_0$ is not periodic when $z_0 \neq 0$.
7. Compute the finite sum $1 + i + i^2 + i^3$ and interpret its geometric cancellation.
8. Describe a real-world process that could be analyzed with at least three of the course's mathematical representations.